BCMS residual sensitivity: background difference and forecast horizon

All test samples · frozen-output removal

Candidate-background difference

(a) Chengdu · inflow

	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	H11	H12
Low n = 221.8k–229.0k	+4.449	+4.659	+5.791	+4.394	+2.251	+3.450	+3.841	+4.549	+5.322	+6.802	+5.050	+4.235
Medium n = 135.9k–138.8k	+2.294	+2.494	+3.236	+2.415	+1.204	+1.925	+2.107	+2.604	+2.995	+3.894	+2.801	+2.343
High n = 135.3k–138.6k	+0.859	+0.972	+1.352	+1.141	+0.518	+0.829	+0.852	+1.081	+1.230	+1.554	+1.104	+0.838

(b) Chengdu · outflow

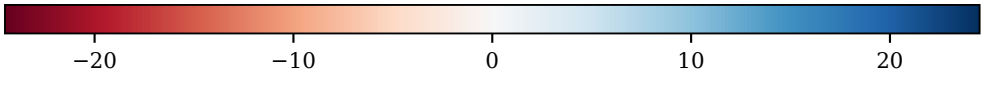
	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	H11	H12
Low n = 244.6k–255.1k	+2.978	+2.525	+1.520	+2.291	+0.440	+0.767	+1.467	+2.799	+1.353	+3.830	+2.258	+1.090
Medium n = 149.8k–155.0k	+1.630	+1.398	+0.856	+1.108	+0.154	+0.331	+0.841	+1.431	+0.709	+2.162	+1.151	+0.503
High n = 122.2k–127.0k	+0.631	+0.547	+0.280	+0.388	+0.053	+0.148	+0.376	+0.590	+0.314	+0.936	+0.443	+0.216

(c) Shanghai · inflow

	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	H11	H12
Low n = 302.4k–307.7k	+23.924	+24.203	+24.480	+24.246	+23.824	+23.927	+23.944	+23.771	+24.502	+24.231	+24.507	+24.264
Medium n = 189.0k–192.8k	+14.355	+14.466	+14.805	+14.594	+14.147	+14.393	+14.297	+14.434	+14.891	+14.287	+14.758	+14.588
High n = 156.6k–158.7k	+8.564	+8.498	+8.680	+8.842	+8.646	+8.673	+8.553	+8.923	+9.181	+8.529	+8.951	+8.910

(d) Shanghai · outflow

	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10	H11	H12
Low n = 293.9k–301.9k	+18.425	+18.107	+18.027	+18.434	+17.674	+18.145	+17.920	+18.359	+18.244	+18.406	+18.453	+18.875
Medium n = 182.4k–188.0k	+11.621	+11.424	+11.545	+11.611	+11.157	+11.343	+11.442	+11.557	+11.468	+11.403	+11.464	+11.643
High n = 153.8k–158.1k	+7.507	+7.292	+7.386	+7.332	+7.112	+7.258	+7.310	+7.123	+7.190	+7.061	+7.187	+7.139



ΔMAE (×10^{-3}; BCMS residual removed − CMSRN)

Values show 1000 × ΔMAE; divide by 1000 to recover original flow units.
ΔMAE = MAE(prediction − BCMS residual) − MAE(CMSRN prediction).
Frozen-output removal, not a retrained ablation or an isolated background test.
H1–H12: 5–60 min. Blue: helps; red: hurts. Counts span valid pairs across horizons.